

Stress Test Four: Times Fonts and Margins

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1 Physics in Times

Newton's second law $F = ma$ and the energy $E = mc^2$ are set in Times, as many journals do. The Schrödinger equation reads

$$i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}, t) = \left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}, t) \right] \Psi(\mathbf{r}, t). \quad (1)$$

A note in
the mar-
gin.

Maxwell's equations in differential form:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \quad \nabla \cdot \mathbf{B} = 0, \quad (2)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (3)$$

A footnote with maths.¹

- Kinetic energy: $K = \frac{1}{2}mv^2$.
- Momentum: $p = mv$, so $K = p^2/(2m)$.
- Work: $W = \int_a^b F(x) dx$.

Plain text keeps flowing after the list: this sentence contains no mathematics at all and should look exactly like ordinary text, without any change of font, size or spacing. Words such as *emphasis* and **bold** are ordinary formatting, not formulas.

2 Chemistry

Water is H₂O, carbon dioxide is CO₂, and the reaction $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ releases energy. The ion NH₄⁺ carries a charge. Temperatures of 25 °C and concentrations of 0.1 mol L⁻¹ are typical.

¹Here $\hbar = h/2\pi \approx 1.055 \times 10^{-34}$ J s.